

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 PRELIMS**

H2 BIOLOGY**9744/3****Paper 3**

Monday

18th September 2017

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

For Examiners' Use	
Section A	
1	/17
2	/18
3	/15
Total	/50

Conceptual error (CE)	Lack of Keywords (K)	Misreading the question (Q)

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

QUESTION 1

Leigh disease is an inherited neuro-metabolic disorder that affects the central nervous system. As the disease progresses, the muscular system is debilitated throughout the body, as the brain cannot control the contraction of muscles. Leigh disease can be caused by a deficiency of the pyruvate dehydrogenase complex (PDHC), most commonly due to a mutation in the X-linked gene, PDHA1, which codes for the PDHC α -subunit.

A married couple (both of whom are normal individuals) is concerned about their chances of having a child with Leigh disease because the woman's father had the disease.

- (a) Draw a genetic diagram with appropriate symbols to determine the probability that they will have an affected child.

.....[6]

200 couples with the same genotypes as the above-mentioned couple were included in a genetic study of the Leigh disease. They had a total of 322 children with the following phenotypes:

85 normal girls
 70 carrier girls
 89 normal boys
 78 boys with Leigh disease

- (b)** A chi-squared test was carried out to test the significance of the difference between the observed (O) and the expected (E) results.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- (i)** Calculate the χ^2 value. Show your working.

.....[2]

χ^2 value:

Fig. 1

Degrees of Freedom	probability								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28

- (ii)** Fig. 1 shows the table of probabilities. State the probability that the observed results does not differ significantly from the expected results.

.....[1]

- (iii)** State what conclusions may be drawn from the probability found in **(ii)**.

.....

[3]

Leigh disease can also be due to a mutation in the mtDNA (mitochondrial DNA) affecting the MT-ATP6 gene. The mutated allele results in the production of a non-functional subunit in the ATP synthase complex which allows protons to pass through the channel without generating the proton motive force.

- (c)** State whether a mitochondrion containing the mutated allele would have a higher, lower or equal oxygen consumption rate relative to a mitochondrion which has the normal allele.

.....[1]

- (d)** Explain your answer in **(c)**.

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.....[2]

The DNA sequence of the MT-ATP6 gene has remained relatively conserved over a long evolutionary period. It is present in a wide diversity of organisms, ranging from the simplest worms to the great apes. Differences in the nucleotide sequence of the gene in different species can be identified using multiple sequence alignments.

- (e)** Discuss how the comparison of MT-ATP6 DNA nucleotide sequences of different species demonstrates evolutionary change.

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.....[2]

[Q1: 17 marks]

QUESTION 2

Celiac disease is an autoimmune disorder that is caused by an improper immune response to the protein gluten, found in wheat, rye, and barley, that damages the lining of the small intestine. There is no cure for celiac, and the only effective treatment is a gluten-free diet.

Recent studies indicated that harmless intestinal viruses, such as the reovirus, can cause the immune system to overreact to gluten, raising the possibility of such viruses contributing to the development of the disease.

Fig. 2.1 shows the general structure of a reovirus. Unlike Human Immuno-deficiency Viruses (HIV), reoviruses are not retroviruses.

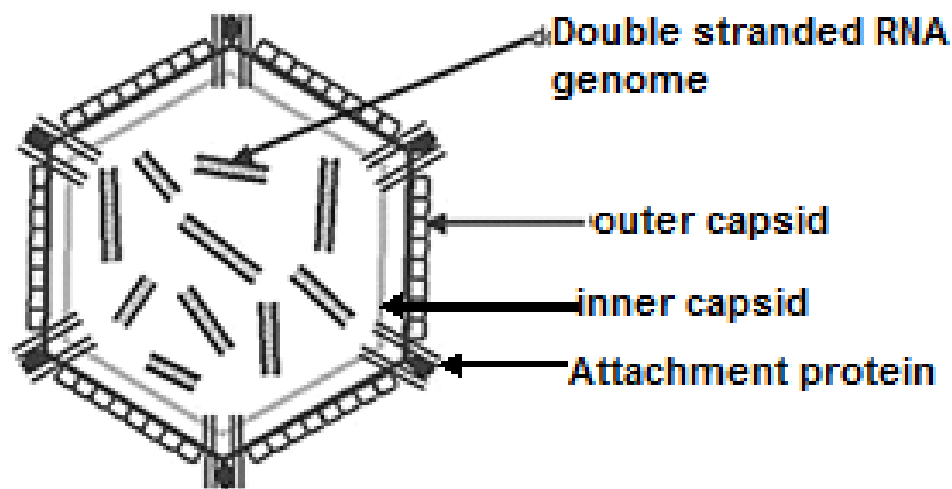


Fig. 2.1

(a) With reference to Fig. 2.1, contrast the structure of the reovirus with that of HIV.

.....

[2]

(b) Suggest how new double-stranded viral RNA genome is synthesized during the reproductive cycle of a reovirus.

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[2]

Researchers also looked at patients with celiac disease and found that they had much higher levels of antibodies against reoviruses than those without the disease.

Those with higher levels of antibodies also had higher levels of the molecule IRF-1 (interferon regulatory factor 1), a regulator of gene transcription which plays a key role in the loss of gluten tolerance.

(c) Describe three ways in which the structure of antibodies contribute to its function.

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.....[3]

Fig. 2.2 shows the structure of an immunoglobulin gene which codes for antibodies.

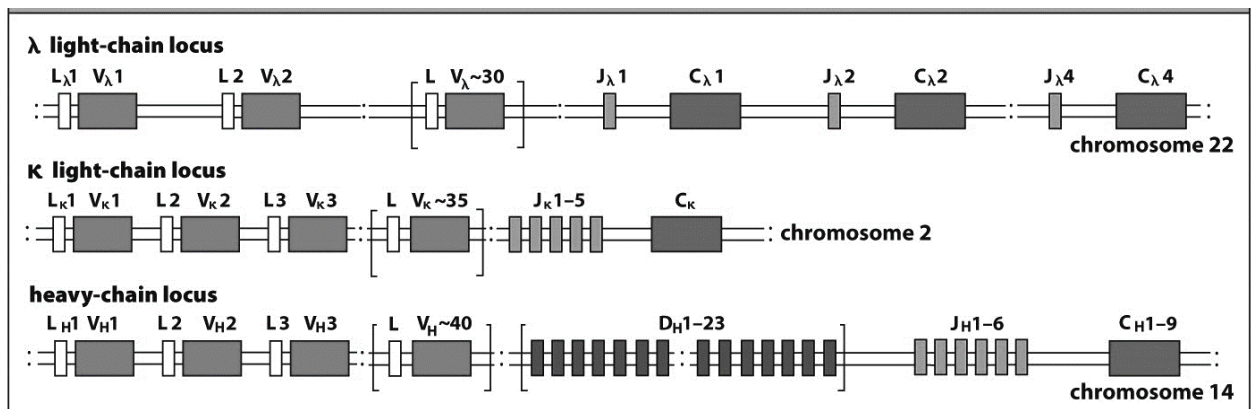


Fig. 2.2

(d) With reference to Fig. 2.2, explain how variability at the DNA level result in variability in the antigen-binding sites of antibodies.

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.....[3]

- (e) Antibodies are proteins. Draw a diagram of the monomer which makes up antibodies. There is no need to annotate the diagram.
[1]

Antibodies are secreted by plasma cells.

Fig. 2.3 shows a plasma cell and a B lymphocyte.

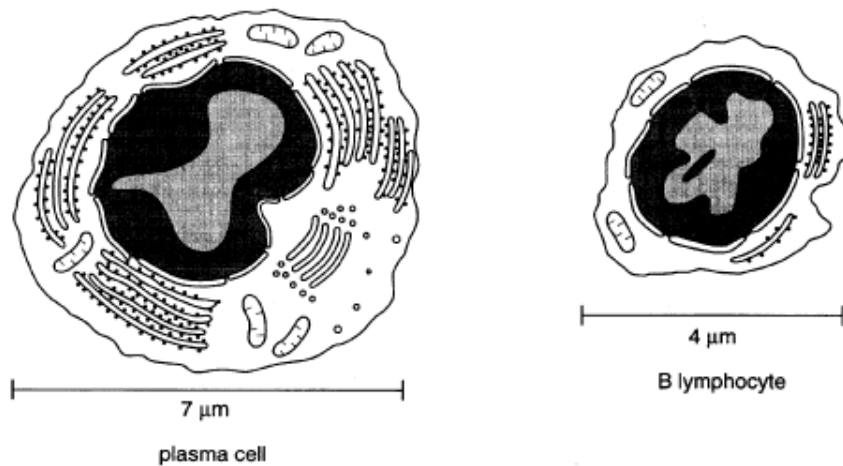


Fig. 2.3

- (f) With reference to Fig. 2.3,
 (i) State 2 ways in which the structure of plasma cell differs from the B lymphocyte.

.....

[2]

- (ii) Explain the reasons for the differences you described.

.....

[2]

A student researcher tried to reproduce the results of the study regarding elevated antibody levels against reoviruses. She studied the levels of antibodies in 3 human subjects infected with reoviruses and another 3 who are not infected.

Table 2.4 shows his results.

Table 2.4

	Antibody titre / unit ml ⁻¹
Subjects not infected with reovirus	30 ± 10
Subjects infected with reovirus	122 ± 25

(g) With reference to the results on subjects not infected with reoviruses, explain what is standard deviation and its implications.

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.....[3]

[Q2: 18 marks]

QUESTION 3

- (a) A student investigated growth in the roots of broad bean, *Vicia faba*. The student cut sections of the root tip of this plant and viewed them with a light microscope.

Fig. 3.1 is a photomicrograph of one of the sections. The cell labelled **D** is in interphase.

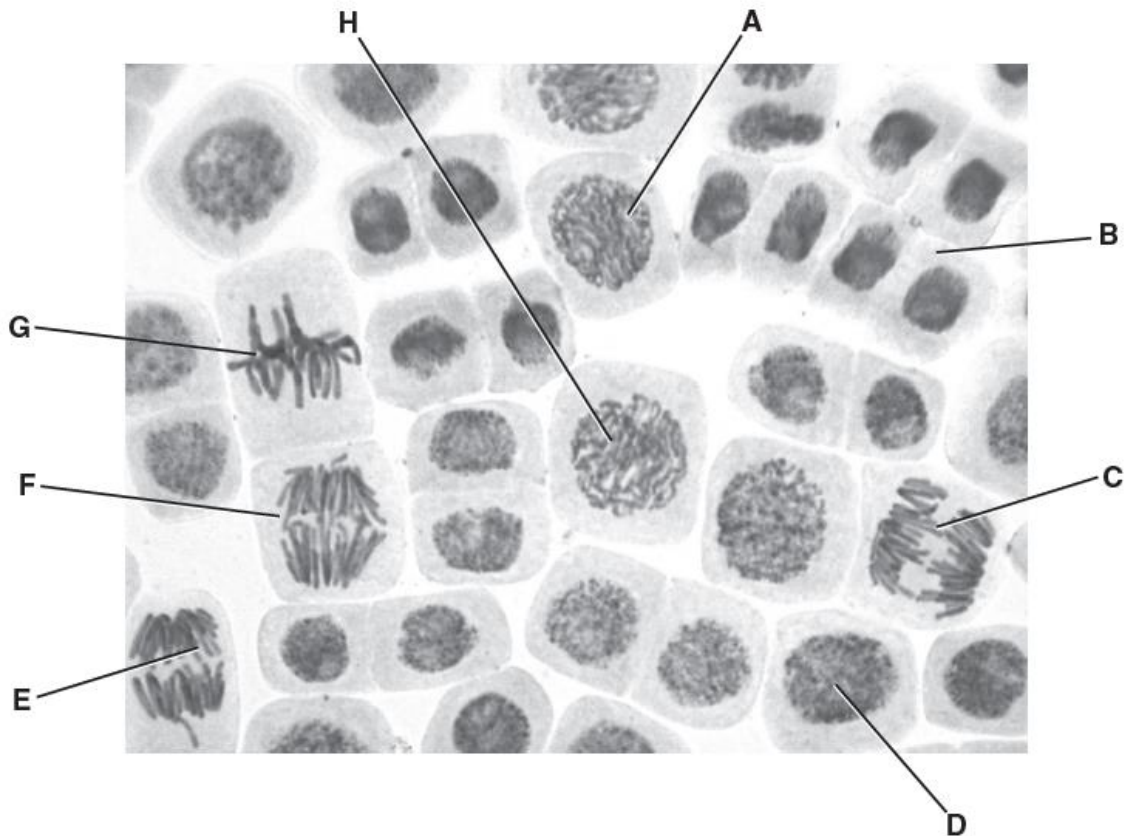


Fig. 3.1

Complete the table below by:

- naming the stages of mitosis in the correct sequence following interphase
- identifying **one** example from the cells labelled **A** to **H** that is in each stage of mitosis that you have named.

.....[3]

stage of mitosis	label from Fig. 3.1

- (b)** In animal cells, centrioles are responsible for assembling microtubules to make the spindle at the beginning of mitosis.

Describe the role of the spindle during mitosis.

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.....[3]

- (c)** Scientists investigated three genes, **P**, **Q** and **R**, involved in controlling cell division. They studied the effect of mutations in these genes on the risk of developing lung cancer.

- (i)** Suggest the differences in the cell cycle of a cancer cell compared with that of a normal cell of the same type.

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.....[2]

The scientists analysed genes, **P**, **Q** and **R** from healthy people and people with lung cancer.

- If a person had at least one copy of normal allele for a gene, they used the symbol N.
- If a person had two mutant alleles for a gene, they used the symbol M.

They used their data to calculate the risk of developing lung cancer for people with different N and M genotypes. A risk value of 1.00 indicates no increased risk. **Table 3.1** shows the scientists' results.

Table 3.1

Gene P	Gene Q	Gene R	Risk value of developing lung cancer / AU
N	N	N	1.00
M	N	N	1.30
N	N	M	1.78
N	M	N	1.45

N = at least one copy of the normal allele is present

M = two copies of the mutant allele are present

- (ii) What do these data suggest about the relative importance of the mutant alleles of genes **P**, **Q** and **R** on **increasing** the risk of developing lung cancer?

.....

[2]

Chemotherapy is the use of a drug to treat cancer. An experiment was set up to study a new drug, SA128, which kills dividing cells. A group of four men suffering from lung cancer was given the drug. The number of cancer cells per unit volume of blood was measured right before treatment and again after the 2-week treatment.

Table 3.2 : Effect of SA128 on cancer cells.

Subject	Number of cancer cells per unit volume of blood					
	TSH	OON	CCK	TTS	Average	Standard deviation
Before drug treatment	2000	1600	1800	1200		
After drug treatment	1000	500	800	300		

- (iii) Calculate the **average** and complete Table 3.2. [1]

- (iv) Using the formula below, calculate the **standard deviations** of number of cancer cells per unit volume of blood before and after drug treatment, SA128. Complete Table 3.2. Express your answers to **3 significant figures**.

.....[1]

$$\text{standard deviation} \quad s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Legend

$\sum$ is 'Sum of'

x is observation

$\bar{x}$ is the mean

n is the sample size (number of observations)

- (v) Using the formula below as well as the average and standard deviation values calculated in (c)(iii) and (c)(iv),

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}}$$

Where:

- $\bar{x}_1$ is the mean of sample 1
- s_1 is the standard deviation of sample 1
- n_1 is the sample size of sample 1
- $\bar{x}_2$ is the mean of sample 2
- s_2 is the standard deviation of sample 2
- n_2 is the sample size in sample 2

Calculate the $t_{\text{calculated}}$ value.

.....[1]

- (vi) Using the critical t-values in **Table 3.3**, determine if there is a significant difference in the results using the new drug, SA128.

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.....[2]

Table 3.3: t-test table of t critical values

df	.10	.05
1	3.078	6.314
2	1.886	2.920
3	1.638	2.353
4	1.533	2.132
5	1.476	2.015
6	1.440	1.943
7	1.415	1.895
8	1.397	1.860
9	1.383	1.833
10	1.372	1.812
11	1.363	1.796
12	1.356	1.782
13	1.350	1.771
14	1.345	1.761
15	1.341	1.753
16	1.337	1.746
17	1.333	1.740
18	1.330	1.734
19	1.328	1.729
20	1.325	1.725
21	1.323	1.721
22	1.321	1.717
23	1.319	1.714
24	1.318	1.711
25	1.316	1.708
26	1.315	1.706
27	1.314	1.703
28	1.313	1.701
29	1.311	1.699
30	1.310	1.697
40	1.303	1.684
60	1.296	1.671
120	1.289	1.658
c	1.282	1.645

[Q3 Total: 15]

Section B

Answer **one** question in this section.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 4 (a)** The effects of global warming on the spread of malaria beyond the tropics is a debatable issue. Discuss the arguments/evidences that support your stance on the matter, and provide a balanced account of counter-arguments. [12]

- (b)** Discuss the effects of climate change (as a result of greenhouse gas emissions). [13]

[Q4 Total: 25]

- 5 (a)** Explain what is meant by primary, secondary, tertiary and quaternary structure of haemoglobin. [13]

- (b)** Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [12]

[Q5 Total: 25]

END